Appendix

Hanwen Liu

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The usage of AI

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Screenshots

R

```r
# Install if you haven't: install.packages("lubridate")
library(lubridate)

# Your vector of messy, mixed-format dates
messy_dates <- c("2023-10-01", "10/02/2023", "03-Oct-2023", "20231004")

# Tell lubridate the possible orders of Year (y), Month (m), and Day (d)
parsed_dates <- parse_date_time(messy_dates, orders = c("ymd", "mdy", "dmy"))

# convert from POSIXct (datetime) to a clean Date format
clean_dates <- as.Date(parsed_dates)

print(clean_dates)
# Output: "2023-10-01" "2023-10-02" "2023-10-03" "2023-10-04"
```

R

```r
# Build the plot layer by layer
ggplot(data = mtcars, aes(x = wt, y = mpg)) +
  # Layer 1: Add the scatter plot points
  geom_point(color = "darkblue", size = 3, alpha = 0.7) +

  # Layer 2: Add the linear regression line and confidence interval
  geom_smooth(method = "lm", color = "red", fill = "lightblue") +

  # Layer 3: Add clean, descriptive labels
  labs(
    title = "Effect of Car Weight on Fuel Efficiency",
    subtitle = "Based on the mtcars dataset",
    x = "Weight (1000 lbs)",
    y = "Miles Per Gallon (MPG)",
    caption = "Linear Regression Line with 95% Confidence Interval"
  ) +

  # Layer 4: Apply a clean, professional theme
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14)
  )
```

Dataset Description

The cleaned dataset consists of `r nrow(clean_data)` observations and `r ncol(clean_data)` variables, representing individual customers engaged during the campaign. The dataset encompasses multiple variable types that provide a comprehensive view of customer behavior. Specifically, it includes continuous numeric variables tracking financial metrics (`marketing_spend`, `purchase_amount`) and engagement (`visits`), categorical variables defining market groups (`segment`), and standardized datetime structures for registration (`signup_date`).

To provide broader context for our analysis, it is essential to understand the composition of our customer base. As illustrated in @fig-segment-counts, our customers are categorized into distinct business segments (e.g., Consumer, Enterprise, SMB). Understanding this distribution ensures that our marketing insights are grounded in the reality of our audience structure. Furthermore, the summary statistics table reveals the average and maximum financial figures across these segments. By examining these aggregated summaries, we can establish a baseline of typical spending behavior before diving into the individual correlational analysis.